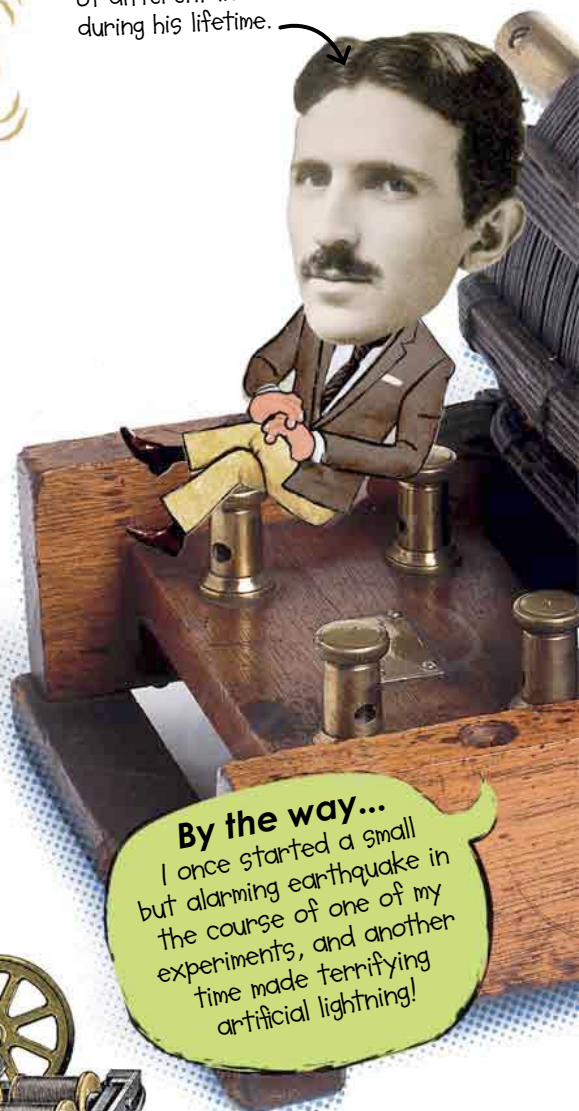


Electric motor

Electric motors use magnetism to produce movement. Today, they are the driving force behind many everyday devices.

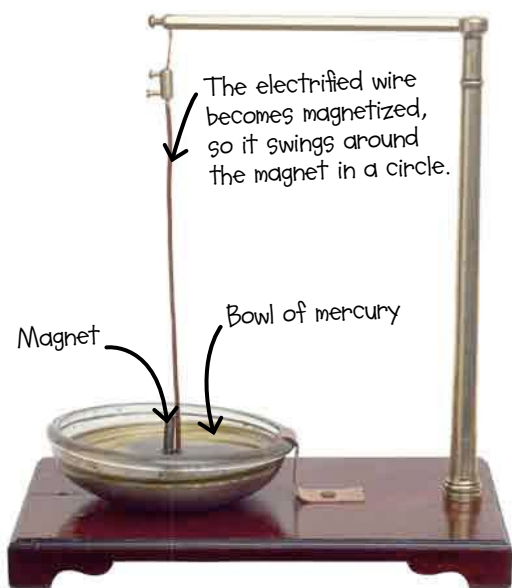
Getting the modern world MOVING

Nikola Tesla was an American engineer who worked on a large number of different inventions during his lifetime.



By the way...

I once started a small but alarming earthquake in the course of one of my experiments, and another time made terrifying artificial lightning!

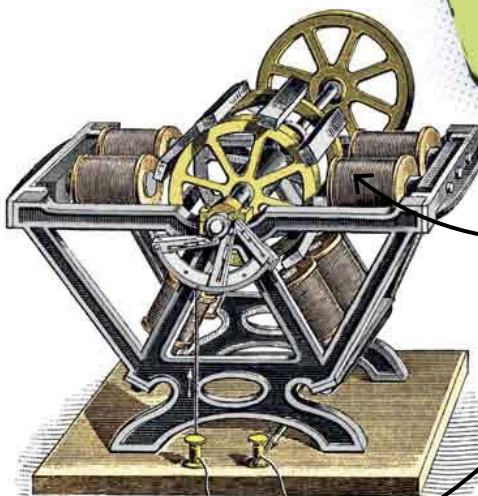


Faraday's electrical experiments

English scientist Michael Faraday made the **first electric motor** in 1821 when he produced **CONTINUOUS MOTION** from electricity. It worked because passing an **electric current** through a wire produces magnetism. Later motors used electromagnets—coils of wire around an iron core—to make this effect stronger.

Motoring on

German engineer **MORITZ VON JACOBI** used electromagnets to make a motor powerful enough to be put to **practical use**. In a world first, an improved version of his motor drove a **paddleboat** across the Neva River in Russia in 1838 with 14 people on board.



Electric current provided by a battery travels through the wire.

It couldn't have happened without...



In 1820, **HANS OERSTED** discovered **ELECTROMAGNETISM**—he found that an electric current could create a magnetic field.



Also in 1820, **ANDRÉ-MARIE AMPÈRE** worked out the relationship between the electromagnetic force and the electric current.